

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Department: CSE - AIDS	Course Code: DSA0604
Course Name: Data Handling and Visualization	

DATA VISUALIZATION — SET 1 TO 5

SET 1: MONTHLY SALES DATA

Question 1

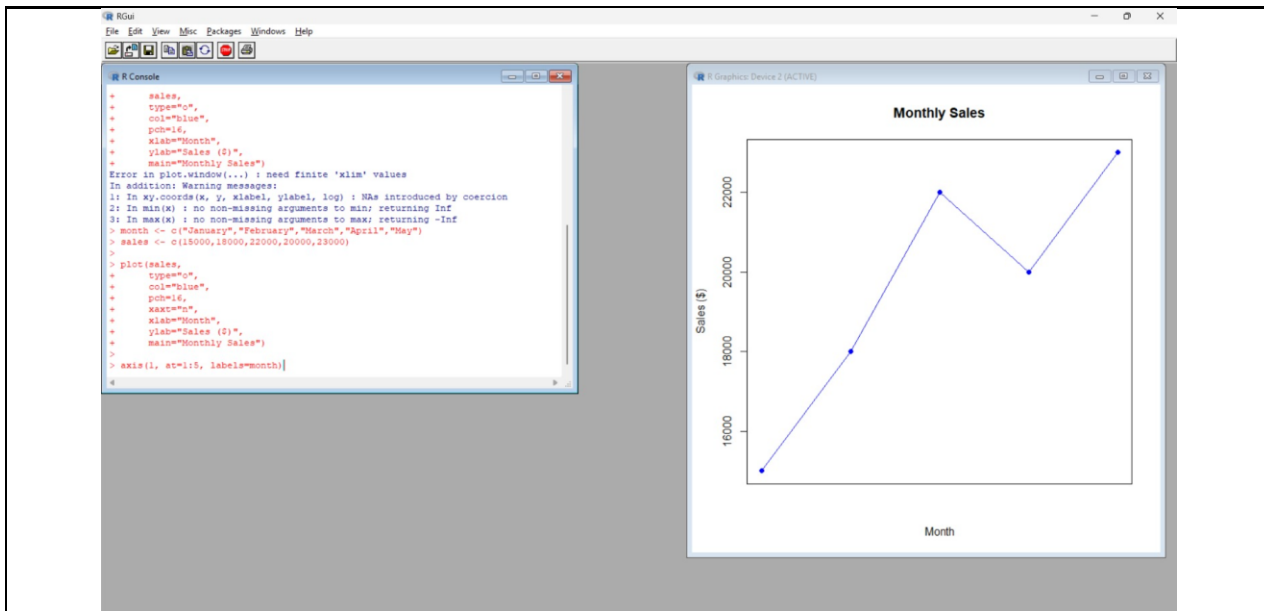
CODE:

```
library(ggplot2)

sales_data <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

ggplot(sales_data, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Sales",
    x = "Month",
    y = "Sales ($)"
  ) +
  theme_minimal()
```

OUTPUT:



Question 2

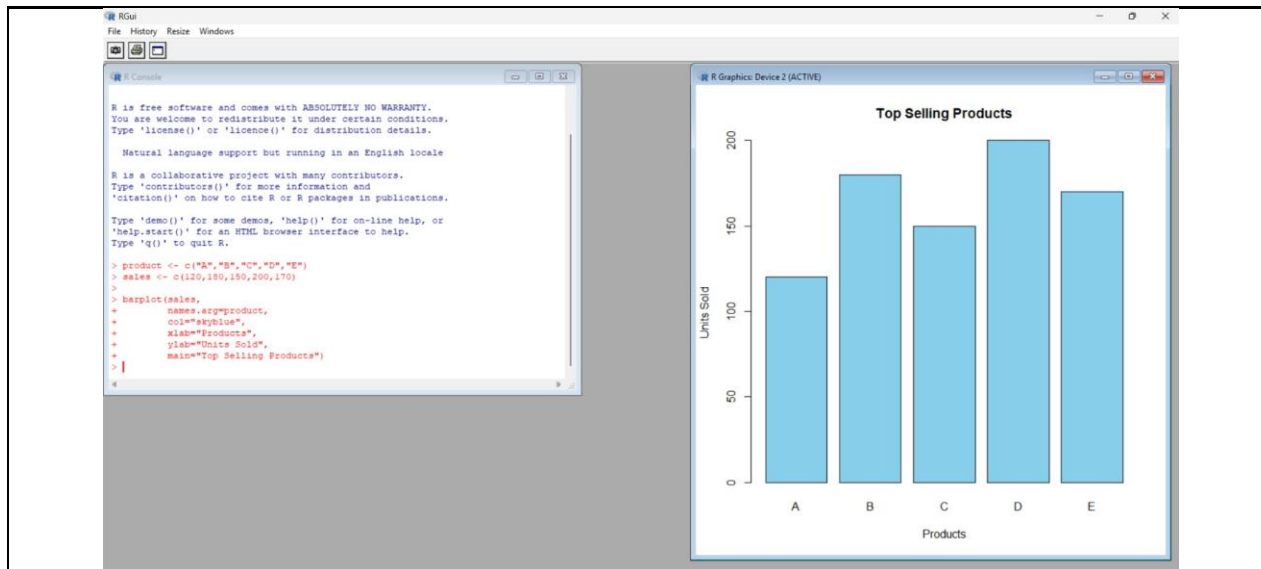
CODE:

```
library(ggplot2)

product_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Sales = c(45000, 38000, 52000, 41000, 47000)
)

ggplot(product_data, aes(x = Product, y = Sales, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top-Selling Products",
    x = "Products",
    y = "Annual Sales ($)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Question 3

CODE:

```

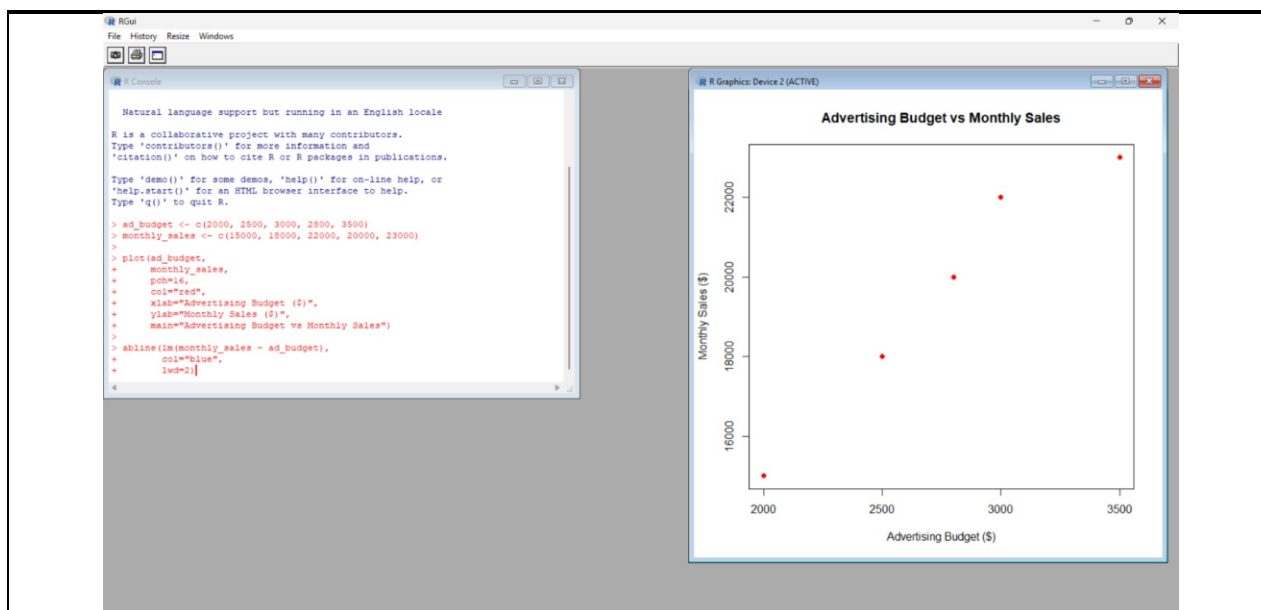
library(ggplot2)

sales_data <- data.frame(
  Advertising_Budget = c(2000, 2500, 3000, 2800, 3500),
  Monthly_Sales = c(15000, 18000, 22000, 20000, 23000)
)

ggplot(sales_data, aes(x = Advertising_Budget, y = Monthly_Sales)) +
  geom_point(color = "blue", size = 4) +
  labs(
    title = "Advertising Budget vs Monthly Sales",
    x = "Advertising Budget ($)",
    y = "Monthly Sales ($)"
  ) +
  theme_minimal()

```

OUTPUT:



Question 4

CODE:

```
library(shiny)
library(ggplot2)

sales_data <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

product_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Sales = c(45000, 38000, 52000, 41000, 47000)
)

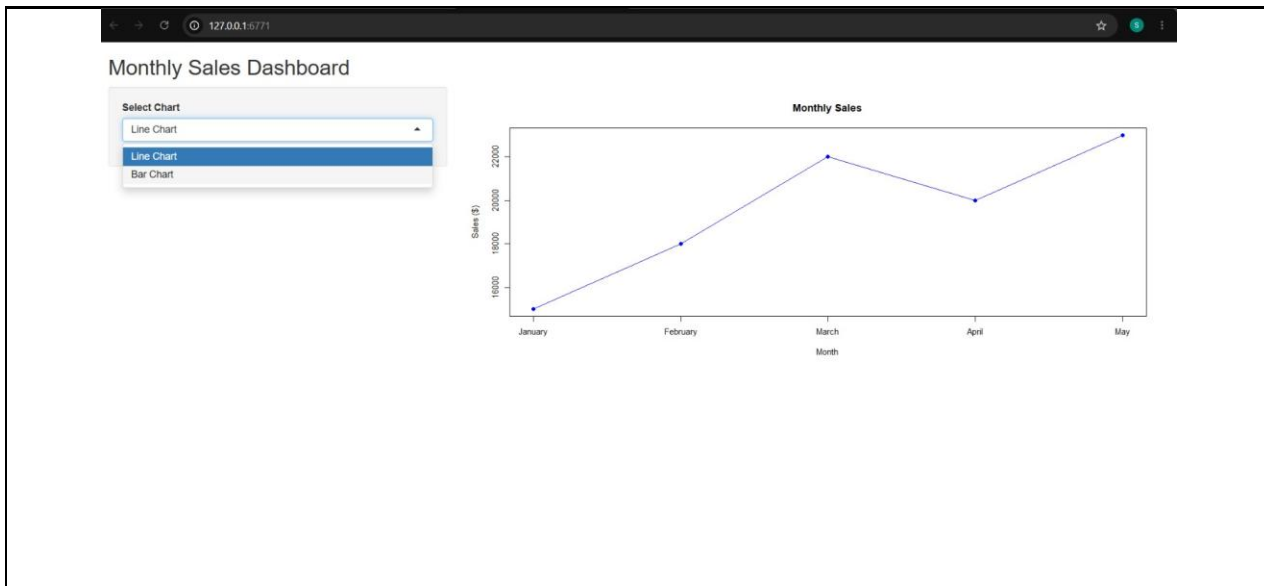
ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  )
)

server <- function(input, output) {
  output$linePlot <- renderPlot({
    ggplot(sales_data, aes(Month, Sales, group = 1)) +
      geom_line(color = "blue", linewidth = 1.2) +
      geom_point(color = "red", size = 3) +
      labs(title = "Monthly Sales", x = "Month", y = "Sales ($)") +
      theme_minimal()
  })

  output$barPlot <- renderPlot({
    ggplot(product_data, aes(Product, Sales, fill = Product)) +
      geom_bar(stat = "identity") +
      labs(title = "Top-Selling Products", x = "Product", y = "Sales ($)") +
      theme_minimal() +
      theme(legend.position = "none")
  })
}

shinyApp(ui, server)
```

OUTPUT:



SET 2: CUSTOMER FEEDBACK ANALYSIS

Question 1

CODE:

```
library(ggplot2)

customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5)
)

ggplot(customer_data, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Frequency"
  ) +
  theme_minimal()
```

OUTPUT:

```

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'Citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> customer_data <- data.frame(
+   CustomerID = c(1, 2, 3, 4, 5),
+   Age = c(25, 30, 35, 28, 40),
+   SatisfactionScore = c(4, 5, 3, 4, 5)
+ )
> hist(
+   customer_data$Age,
+   main = "Distribution of Customer Ages",
+   xlab = "Age",
+   ylab = "Frequency",
+   col = "skyblue",
+   border = "black",
+   breaks = 5
+ )
>
4

```



Question 2

CODE:

```

library(ggplot2)

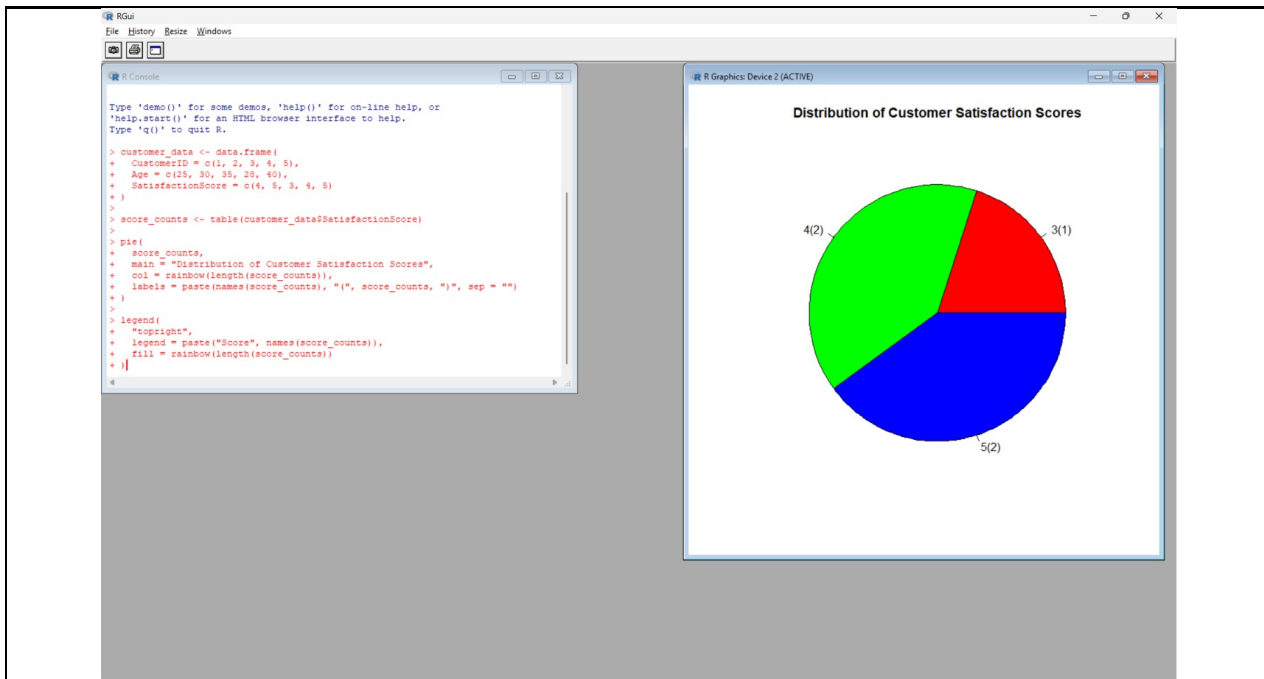
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5)
)

score_data <- as.data.frame(table(customer_data$Satisfaction_Score))
colnames(score_data) <- c("Score", "Count")

ggplot(score_data, aes(x = "", y = Count, fill = Score)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(
    title = "Distribution of Customer Satisfaction Scores",
    fill = "Score"
  ) +
  theme_void()

```

OUTPUT:



Question 3

CODE:

```

library(ggplot2)

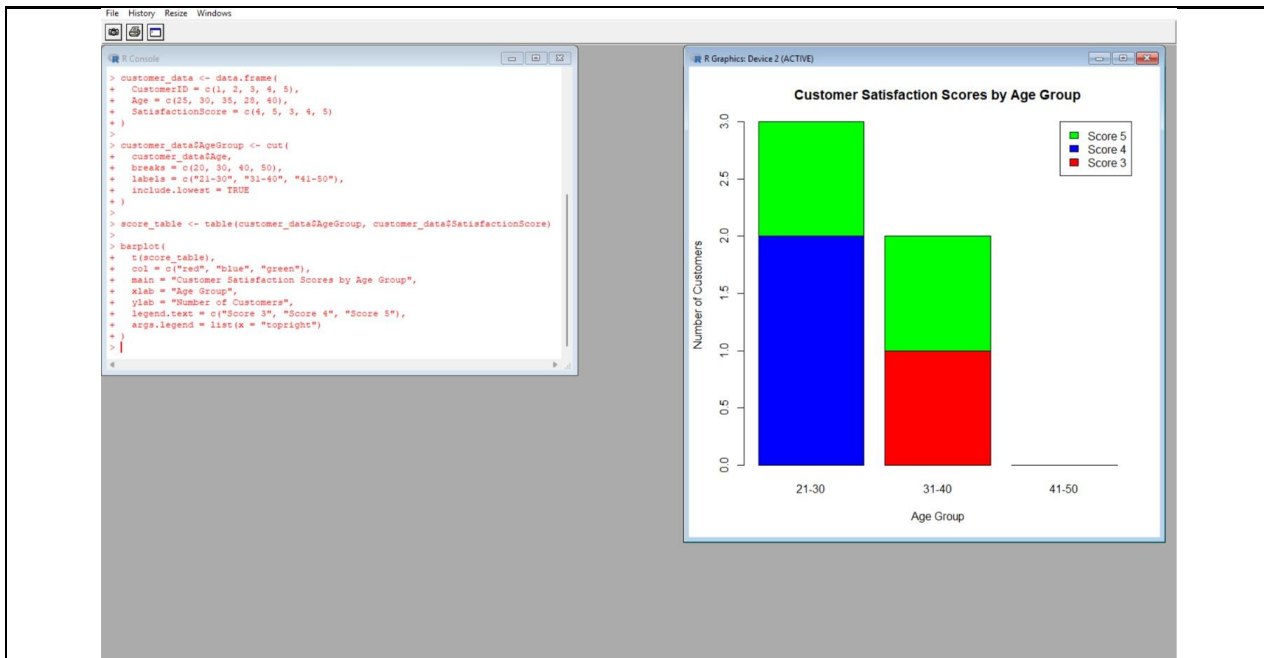
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = factor(c(4, 5, 3, 4, 5))
)

customer_data$Age_Group <- cut(
  customer_data$Age,
  breaks = c(20, 30, 40, 50),
  labels = c("21-30", "31-40", "41-50"),
  right = TRUE
)

ggplot(customer_data, aes(x = Age_Group, fill = Satisfaction_Score)) +
  geom_bar() +
  labs(
    title = "Customer Satisfaction Scores by Age Group",
    x = "Age Group",
    y = "Number of Customers",
    fill = "Satisfaction Score"
  ) +
  theme_minimal()

```

OUTPUT:



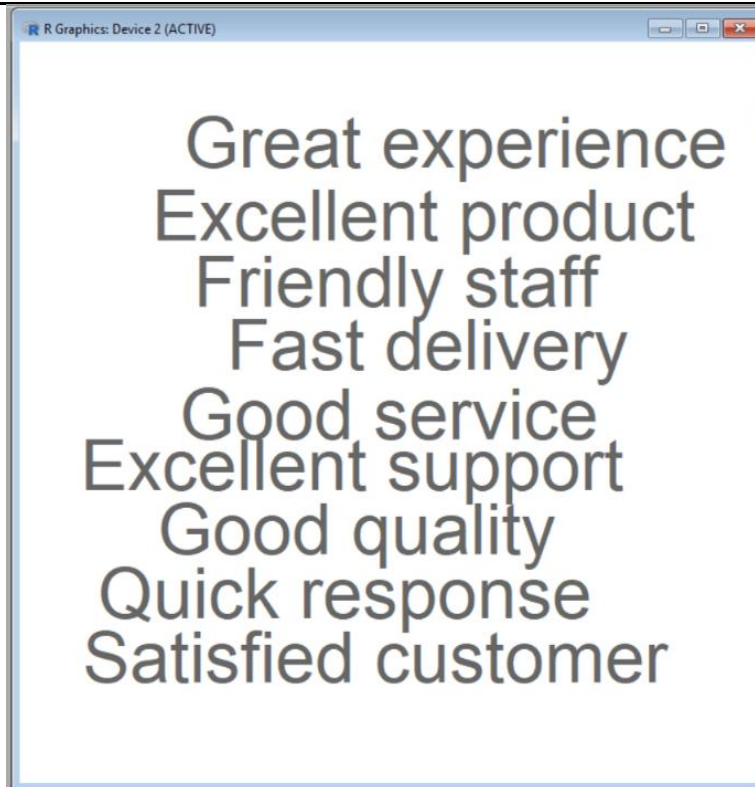
Question 4

CODE:

```
feedback <- c(
  "Good service",
  "Excellent support",
  "Fast delivery",
  "Friendly staff",
  "Good quality",
  "Excellent product",
  "Quick response",
  "Satisfied customer",
  "Great experience",
  "Highly recommended"
)

wordcloud(
  words = feedback,
  freq = rep(1, length(feedback)),
  min.freq = 1,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```

OUTPUT:



SET 3: EMPLOYEE PERFORMANCE EVALUATION

Question 1

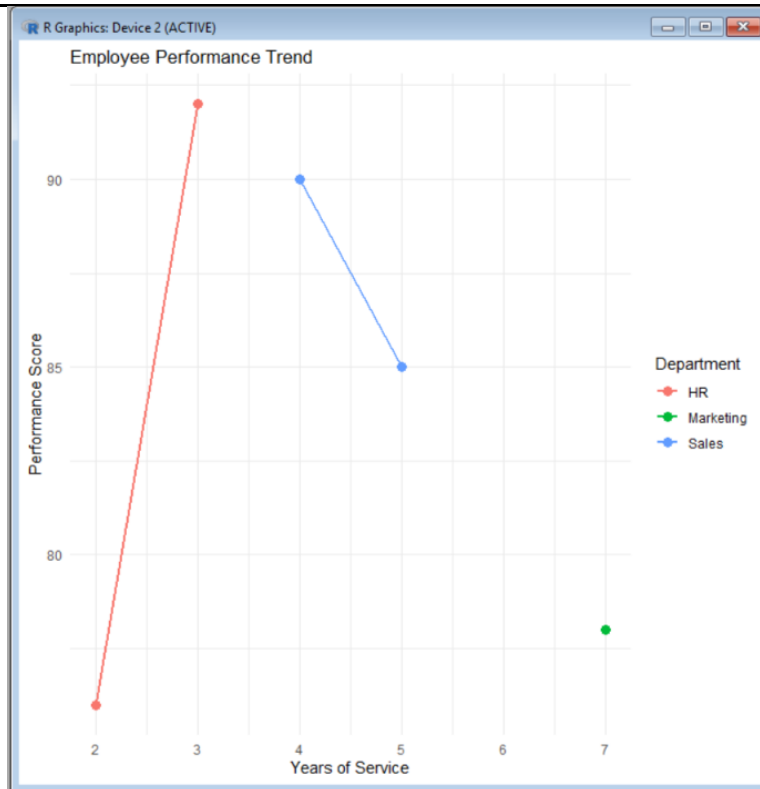
CODE:

```
library(ggplot2)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score, color = Department))
+
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score",
    color = "Department"
  ) +
  theme_minimal()
```

OUTPUT:



Question 2

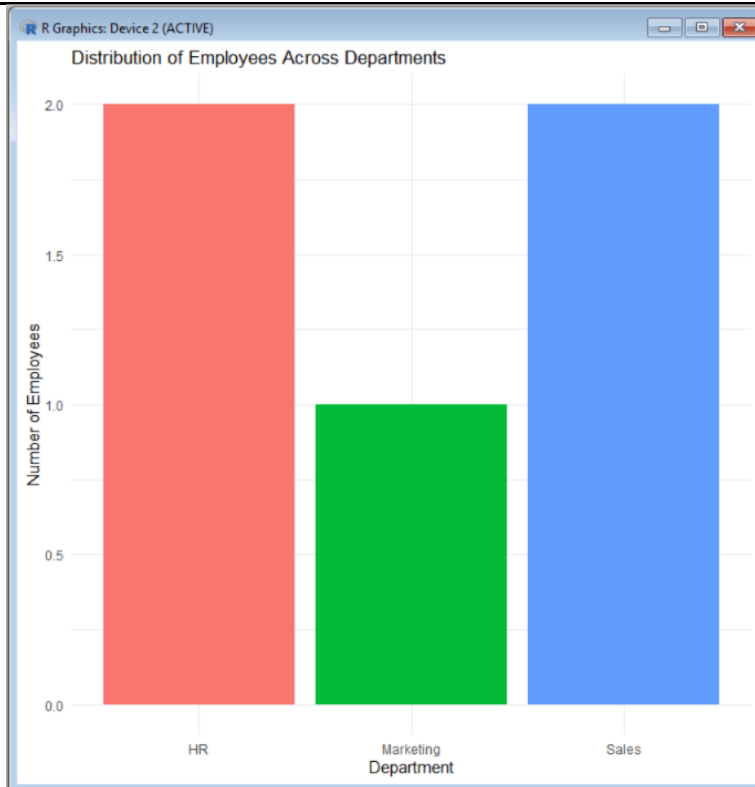
CODE:

```
library(ggplot2)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ggplot(employee_data, aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Distribution of Employees Across Departments",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Question 3

CODE:

```
library(ggplot2)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score, color = Department))
+
  geom_point(size = 4) +
  labs(
    title = "Years of Service vs Performance Score",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

OUTPUT:



Question 4

CODE:

```
library(shiny)
library(ggplot2)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  )
)

server <- function(input, output) {
  output$linePlot <- renderPlot({
    ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score, color =
Department)) +
      geom_line(linewidth = 1) +
      geom_point(size = 3) +
      labs(
        title = "Employee Performance Trend",
        x = "Years of Service",
        y = "Performance Score",
        color = "Department"
      )
  })
}
```

```

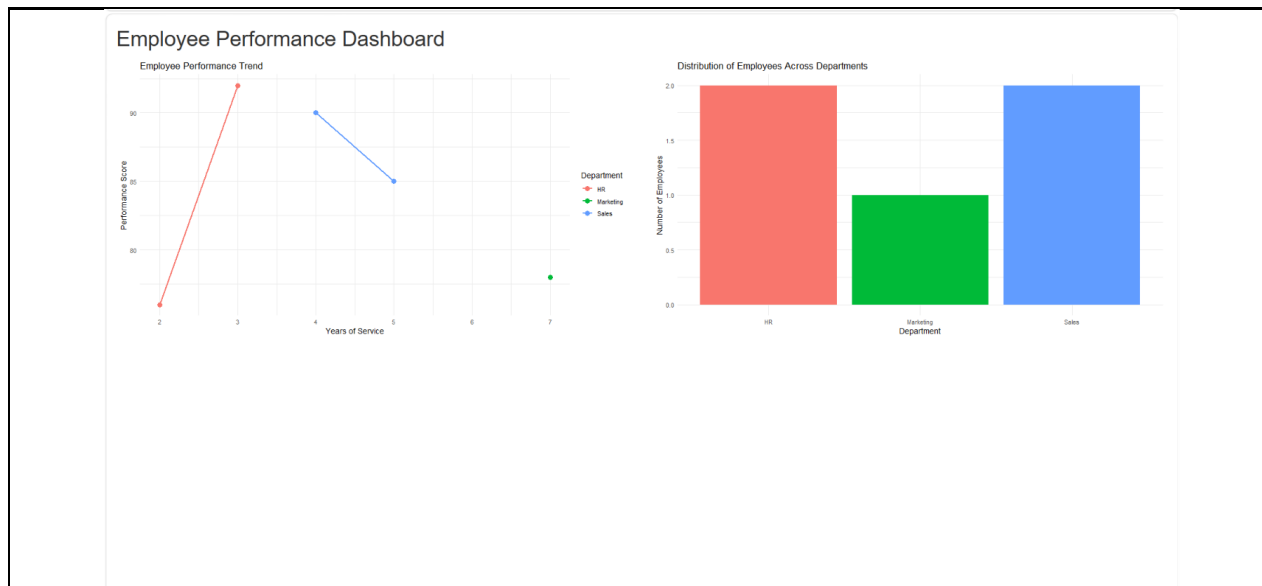
    theme_minimal()
  })

  output$barPlot <- renderPlot({
    ggplot(employee_data, aes(x = Department, fill = Department)) +
      geom_bar() +
      labs(
        title = "Distribution of Employees Across Departments",
        x = "Department",
        y = "Number of Employees"
      ) +
      theme_minimal() +
      theme(legend.position = "none")
  })
}

shinyApp(ui, server)

```

OUTPUT:



SET 4: PRODUCT INVENTORY MANAGEMENT

Question 1

CODE:

```

library(ggplot2)

inventory_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

ggplot(inventory_data, aes(x = Product, y = Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",
    x = "Product",

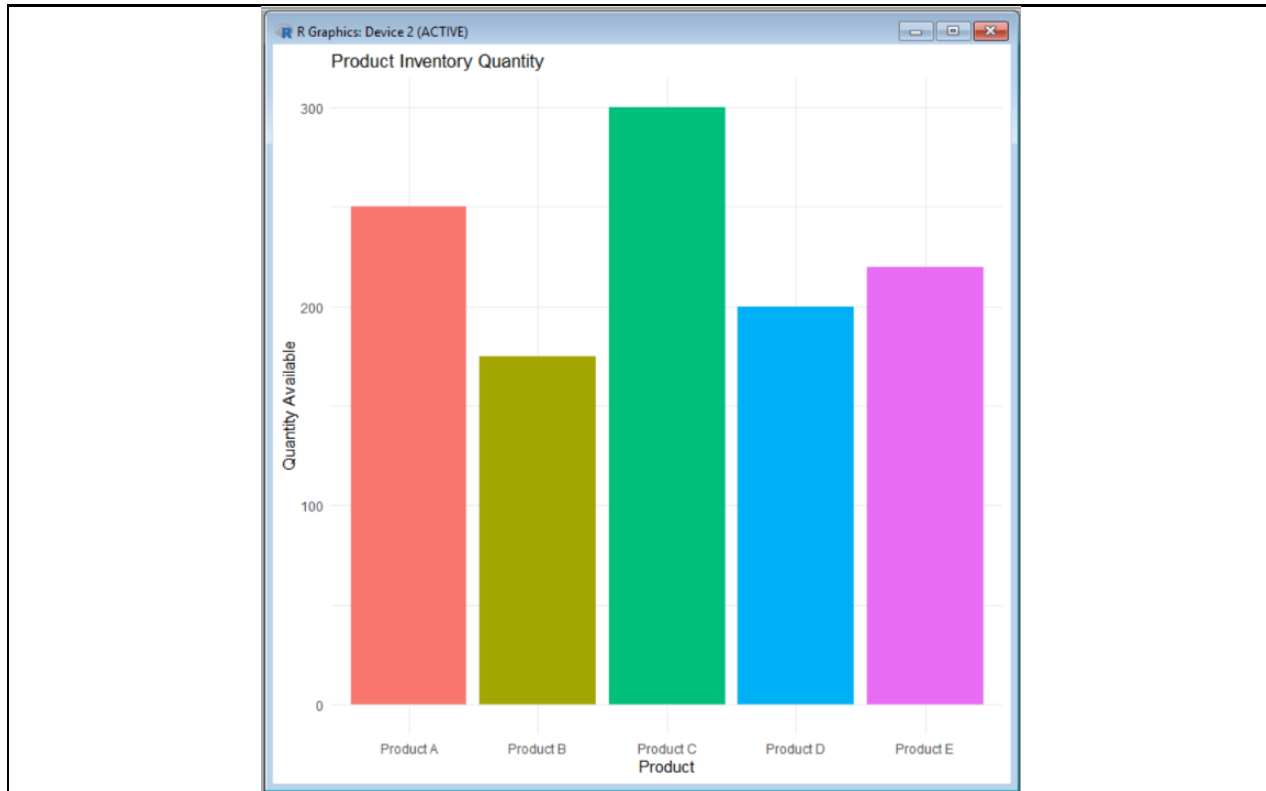
```

```

  y = "Quantity Available"
) +
theme_minimal() +
theme(legend.position = "none")

```

OUTPUT:



Question 2

CODE:

```

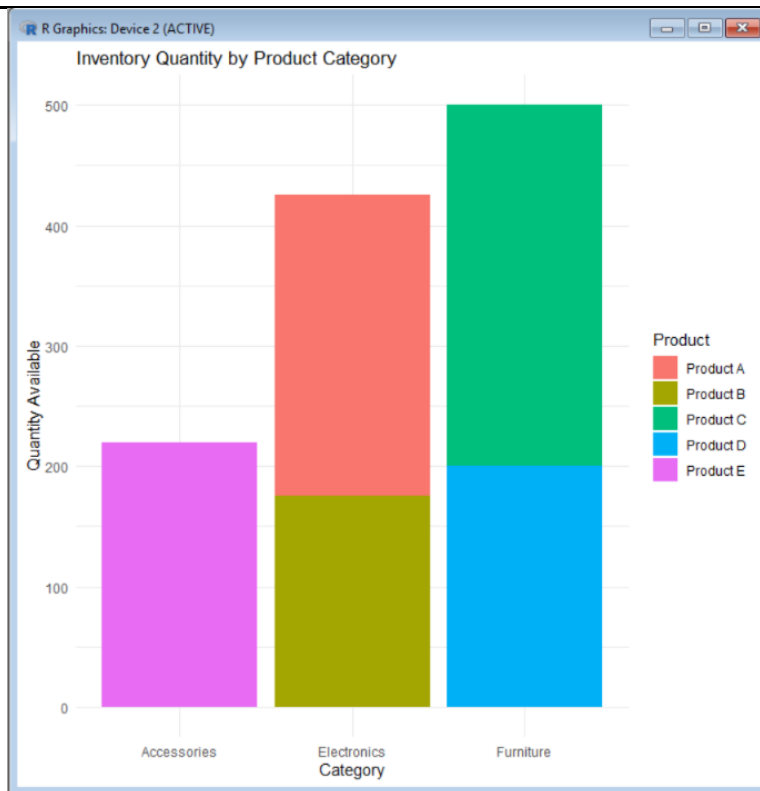
library(ggplot2)

inventory_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories"),
  Quantity = c(250, 175, 300, 200, 220)
)

ggplot(inventory_data, aes(x = Category, y = Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Inventory Quantity by Product Category",
    x = "Category",
    y = "Quantity Available",
    fill = "Product"
  ) +
  theme_minimal()

```

OUTPUT:



Question 3

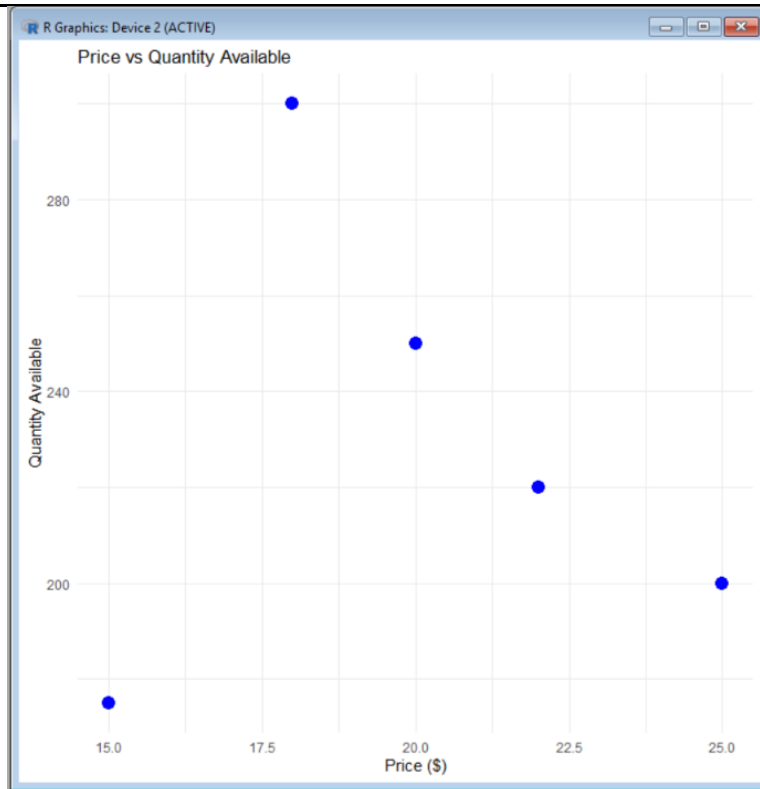
CODE:

```
library(ggplot2)

inventory_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Price = c(20, 15, 18, 25, 22),
  Quantity = c(250, 175, 300, 200, 220)
)

ggplot(inventory_data, aes(x = Price, y = Quantity)) +
  geom_point(size = 4, color = "blue") +
  labs(
    title = "Price vs Quantity Available",
    x = "Price ($)",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

OUTPUT:



Question 4

CODE:

```
library(shiny)
library(ggplot2)

bar_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)

stack_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories"),
  Quantity = c(250, 175, 300, 200, 220)
)

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  fluidRow(
    column(6, plotOutput("barPlot")),
    column(6, plotOutput("stackPlot"))
  )
)

server <- function(input, output) {
  output$barPlot <- renderPlot({
    ggplot(bar_data, aes(Product, Quantity, fill = Product)) +
      geom_bar(stat = "identity") +
      labs(
        title = "Product Inventory Quantity",
        x = "Product",
```



```

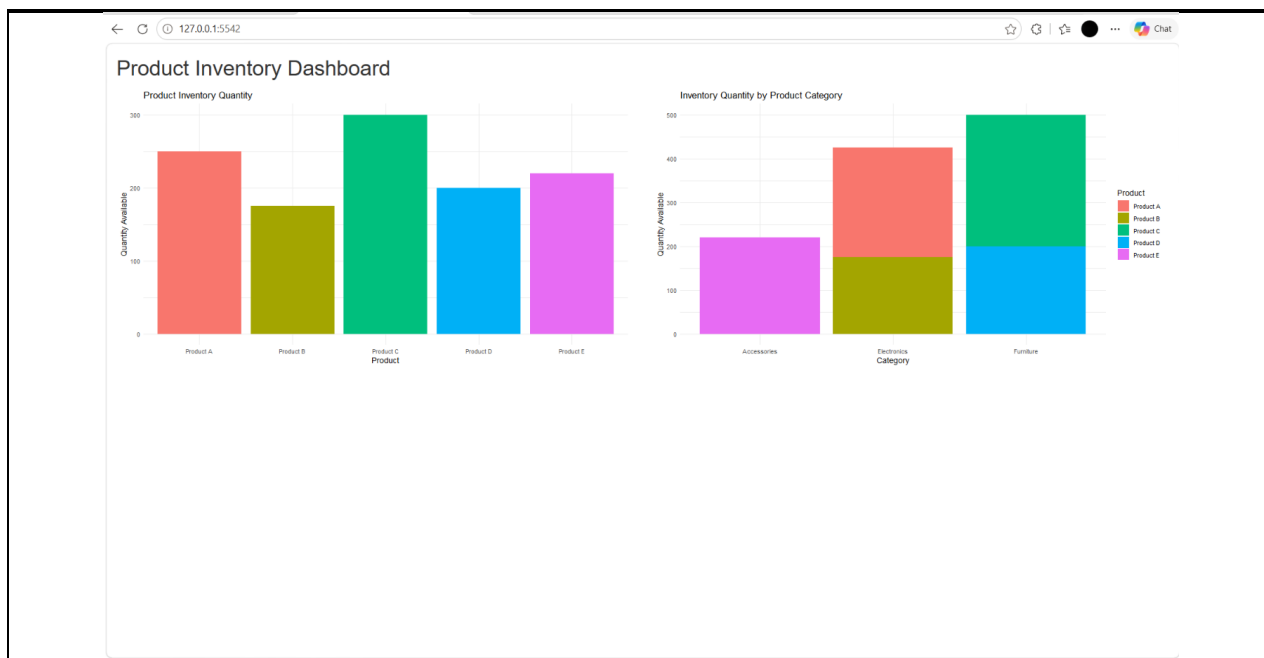
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
})

output$stackPlot <- renderPlot({
  ggplot(stack_data, aes(Category, Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Inventory Quantity by Product Category",
    x = "Category",
    y = "Quantity Available"
  ) +
  theme_minimal()
})
}

shinyApp(ui, server)

```

OUTPUT:



SET 5: WEBSITE ANALYTICS

Question 1

CODE:

```

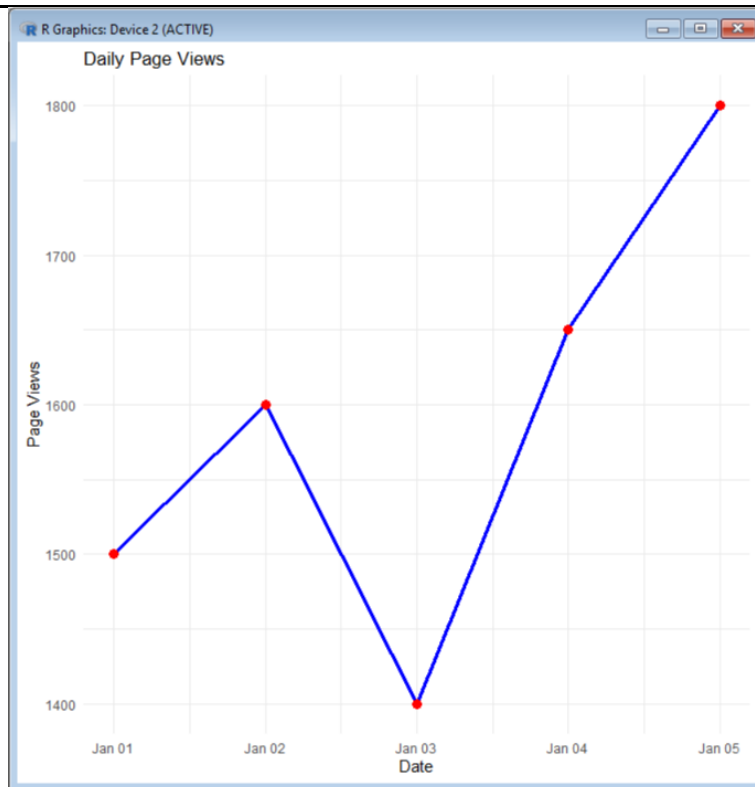
library(ggplot2)

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

```

```
ggplot(website_data, aes(x = Date, y = Page_Views, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()
```

OUTPUT:



Question 2

CODE:

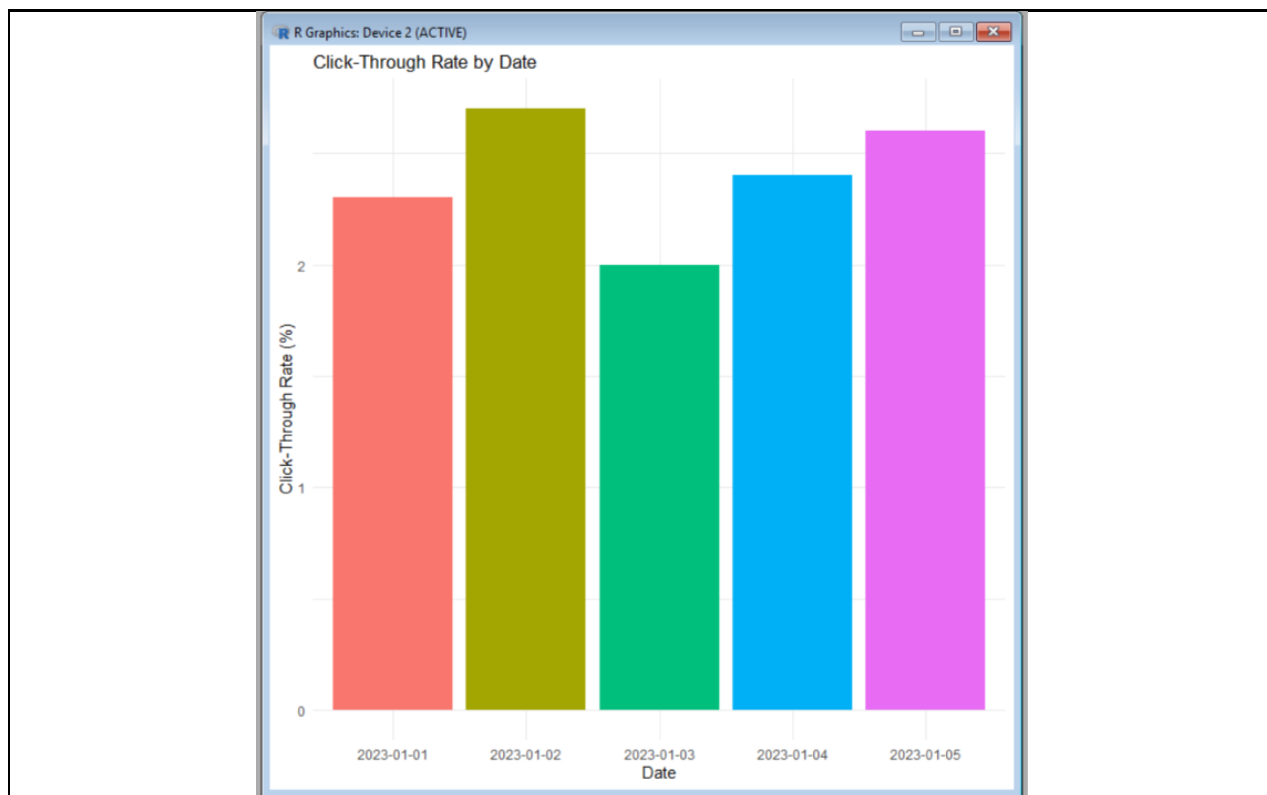
```
library(ggplot2)

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ggplot(website_data, aes(x = factor(Date), y = Click_Through_Rate, fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Click-Through Rate by Date",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal() +
```

```
theme(legend.position = "none")
```

OUTPUT:



Question 3

CODE:

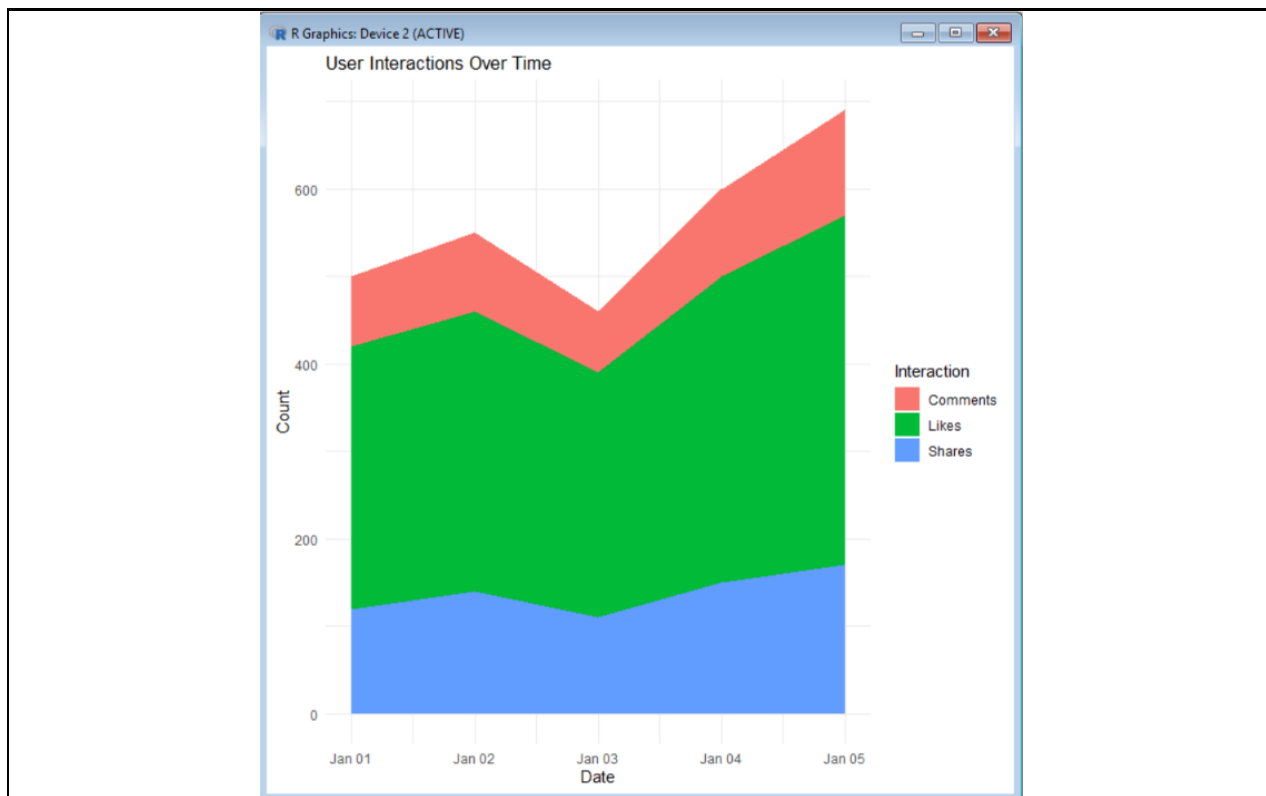
```
library(ggplot2)
library(tidyr)

interaction_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Likes = c(300, 320, 280, 350, 400),
  Shares = c(120, 140, 110, 150, 170),
  Comments = c(80, 90, 70, 100, 120)
)

interaction_long <- pivot_longer(
  interaction_data,
  cols = c(Likes, Shares, Comments),
  names_to = "Interaction",
  values_to = "Count"
)

ggplot(interaction_long, aes(x = Date, y = Count, fill = Interaction)) +
  geom_area() +
  labs(
    title = "User Interactions Over Time",
    x = "Date",
    y = "Count"
  ) +
  theme_minimal()
```

OUTPUT:



Question 4

CODE:

```
library(shiny)
library(ggplot2)

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  )
)

server <- function(input, output) {
  output$linePlot <- renderPlot({
    ggplot(website_data, aes(Date, Page_Views, group = 1)) +
      geom_line(color = "blue", linewidth = 1.2) +
      geom_point(color = "red", size = 3) +
      labs(
        title = "Daily Page Views",
        x = "Date",
        y = "Page Views"
      ) +
      theme_minimal()
  })
}
```

```

}))

output$barPlot <- renderPlot({
  ggplot(website_data, aes(factor(Date), Click_Through_Rate, fill = factor(Date))) +
    geom_bar(stat = "identity") +
    labs(
      title = "Click-Through Rate by Date",
      x = "Date",
      y = "Click-Through Rate (%)"
    ) +
    theme_minimal() +
    theme(legend.position = "none")
})
}

shinyApp(ui, server)

```

OUTPUT:

